

Lab 01: Data wrangling and EDA

A. Executive Summary: What are your main findings and recommendations, in a single paragraph?

While the University of Virginia (UVA) has valuable foreign gift activity, receiving around \$22.2 million across 98 reported foreign monetary gifts and contracts, the university falls behind not only in comparison to institutions receiving the greatest number and number of gifts overall, but also in comparison to similar public peer institutions and other universities located in Virginia. UVA's foreign funding is also concentrated among a relatively small number of countries and sources, with Bangladesh, Sweden, and Tanzania being its largest sources by total dollar amount. Therefore, UVA still has substantial opportunities to expand funding through strategic action. My key recommendations are that UVA focus on identifying foreign sources that already support peer institutions, particularly UNC – Chapel Hill. This approach could allow UVA to target proven sources of foreign funding rather than attempting to develop entirely new markets. A secondary consideration could be for UVA to look more into developing contracts with other countries, as this seems to be the primary source of funding for peer institutions, such as UNC.

B. Summarize the results of your initial analyses of overall giving and giving to UVA (1-3 paragraphs).

Overall, based on my initial analysis, total giving is highly skewed in that the average amount that each university receives is a little over \$52,000,000 but the median amount is around \$6,500,000; this suggests that there are a small number of universities that receive extremely large amounts and dominate the foreign giving space. In addition, the most frequent type of gift given is contracts and the least common gift type, by far, is real estate. Notably, the greatest source of foreign gifts in total comes from Qatar, gifting around \$2.71 billion, followed by England (\$1.46 billion), and then China (\$1.23 billion). England is the country giving the greatest number of gifts, giving 3,655. In analyzing gift size, Bermuda gives the largest average gifts, with an average value of \$7.7 million. The U.S. institution receiving the most money in total is Carnegie Mellon University, who receives close to \$1.5 billion, while UCLA has the largest number of reported transactions at 3,916.

UVA received \$22.2 million through 98 reported foreign transactions. Consistent with the overall trend in the dataset, contracts were the most common type of transactions for UVA, accounting for 64 of the 98 total transactions, with the remaining 34 classified as monetary gifts. UVA's average gift value was approximately \$226,000, while the median was approximately \$80,500, indicating that several relatively large transactions increase UVA's average gift value.

C. Describe a strategy for finding new potential sources of money for UVA. What countries or institutions should UVA target, and why? (1-3 paragraphs)

UVA should focus first on foreign sources that already provide funding to comparable

universities, rather than attempting to start from scratch elsewhere. UNC – Chapel Hill is a particularly useful target because it has a similar academic profile and receives approximately \$41.3 million in foreign funding compared with UVA's \$22.2 million. UNC also has 286 reported transactions compared with UVA's 98. Importantly, UNC's advantage appears to come primarily from contracts: 267 of its 286 transactions are contracts, compared with 64 of UVA's 98 transactions. Interestingly, UVA actually has more monetary-gift transactions than UNC (34 versus 19). One potential strategy is to identify countries and sources that already give to UNC but either do not give to UVA or give less to UVA. There are five countries that give money to both universities but provide more to UNC: England, France, Japan, Switzerland, and the Netherlands. Switzerland is a particularly notable difference, with UNC receiving approximately \$5.4 million and UVA receiving around \$2.0 million. France is another opportunity, with UNC receiving approximately \$2.6 million compared with only \$250,000 for UVA. These are promising targets, as these countries have already established an interest in giving money to UVA and UVA would not need to establish an entirely new relationship.

UVA could also investigate countries that provide substantial funding to UNC but currently have no reported funding to UVA, including Canada (\$14.0 million to UNC), Zambia (\$4.35 million), and Taiwan (\$2.08 million). These countries should be viewed as places to investigate further to determine what organizations are funding UNC and whether UVA has comparable academic programs and research capabilities.

In addition to UNC, UVA can also learn from other Virginia institutions, such as George Mason, George Washington, and VCU. These universities receive substantially more foreign funding than UVA; fascinatingly, much of this difference comes from contracts rather than monetary gifts. For instance, GMU's \$65.4 million and VCU's \$89.2 million in foreign funding consist entirely of reported contracts, while UVA has both contracts and monetary gifts. While this should not be the primary strategy goal for UVA to increase foreign funding, as it may prove a more difficult approach, UVA can potentially use these peer, Virginia institutions as providing a useful model for developing this form of foreign partnerships through contracts in the future.

D. Conclude with a summary of UVA's prospects for increasing foreign giving. (1 paragraph).

In summation, UVA has significant opportunities to increase its foreign funding to ultimately become a more competitive player in the space of foreign gifts. An encouraging strategy is to pursue proven sources rather than simply targeting countries with large amounts of foreign wealth. The most actionable opportunity is for UVA to identify foreign sources that currently support UNC and other peer institutions but have limited relationships with UVA, such as Switzerland, France, England, Japan, and the Netherlands, as well as countries where UNC has substantial funding that UVA does not. UVA has already developed a solid foundation of funding and has a relatively high number of monetary gift transactions compared with other Virginia institutions; now, UVA just needs to expand upon this.